

Approximating Infinite Impulse Response Filters with Finite Impulse Response Filters Using Allan Variance^{*}

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Abstract: This work presents the application of Allan VARIance (AVAR) to determine the order of an FIR filter whose output approximates that of an IIR filter. AVAR methods are typically used to analyze the variance of static windowed averages of data; prior recent work by the authors extended AVAR methods to include optimization of moving average filter designs, and more recently optimization of finite impulse response (FIR) filters. One of the main advantages of FIR over IIR filters is that the output of FIR depends only on the data size equal to one more than the filter order, whereas the data size influencing the output of the IIR filter increases with the length of data. A consequence of this is that the AVAR optimization of IIR filters remains an unsolved problem, but one that is solvable if IIR filters can be well approximated by FIR filter designs. In this work, the similarity between FIR and IIR filters is quantified using the normalized distance between the AVAR curves of error. This approximation is demonstrated through time-domain results of a signal with both low- and high-frequency noise contributions, namely drift (random walk) input corrupted by white noise. The results show that the AVAR-equivalent FIR filter output is similar to that of the IIR filter output if one chooses a sufficiently high FIR filter order. In addition, an iterative algorithm is presented to quickly estimate the necessary order of an FIR filter that is AVAR-equivalent to an IIR filter.

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1. INTRODUCTION

Allan VARIance (AVAR) is a technique first developed to study the frequency stability of atomic clocks by Allan (1966) and Barnes (1971) at the National Bureau of Standards, Boulder, Colorado for noise types including white noise phase modulation, flicker noise phase modulation, white noise frequency modulation, and flicker noise frequency modulation. The advantage of AVAR over standard variance is its invariability to data length. Howe et al. (1981) presented confidence bounds for AVAR estimates and empirical formulae to calculate degrees of freedom of AVAR estimate. Galleani and Tavella developed Dynamic Allan VARIance (DAVAR) to study changing clock behavior in the Galileo satellite system in a series of publications Galleani and Tavella (2003, 2005); Galleani and Tavella (2009); Galleani (2010, 2011). AVAR applications include quantifying the stability of clocks Tetu et al. (1973); Busca et al. (1975), lasers Kartaschoff and Barnes (1972); Lerberghe et al. (1978); Hori et al. (1983), interferometers Rogers and Moran (1981), oscillators Giles et al. (1989), and the characterization of inertial sensors Li and Fang (2013); Bhardwaj et al. (2015). While AVAR methods his-

torically focused on hardware and sensor characterization, these same methods readily extend to signal processing and controller design. For example, one recent development uses AVAR to develop an optimal moving average filter for filtering white noise, see Haeri et al. (2021).

This paper is motivated by a project requiring real-time signal analysis for autonomous vehicles aggregating information in space and time via 2-dimensional AVAR. For example, our vehicle moving at a highway speed of $25 \frac{m}{s}$ at a sampling frequency of $500 Hz$ generates 20,000 measurement data points per kilometer, per instrumented tire. Fast-AVAR (FAVAR) methods were developed in Maddipatla et al. (2021) for real-time (less than 0.5s) estimation of massive quantities of road-friction data. This prior work used raw data samples, but the need to extend results to MA, FIR, and IIR filtering were observed which led to the contributions herein. In extending AVAR from the typical application areas of static data analysis to dynamic systems, one can observe that AVAR is an analysis of changing averages. The closest analogy within classical input/output signal processing is the MA filter, wherein averages are calculated along a moving window of a signal. The results of Maddipatla and Brennan (2024) extended AVAR analysis to MA filters. A concurrent paper (Maddipatla and Brennan (2025)) extends the optimization of MA filters to include the general form of FIR filters, namely, optimization of a moving-average type filter with non-

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uniform coefficients. This work extends the prior AVAR analysis of MA and FIR filters to include the general form of IIR filters. In addition, it presents an AVAR based algorithm to find the order of FIR filter that produces a response that is equivalent, within an epsilon bound of an AVAR curve difference, to an IIR filter - hereafter called AVAR-equivalent.

To motivate the problem, the definition and notation of signals is helpful. Figure 1 shows the system diagram of the IIR filter. Here, $r(k)$ is the reference or true input, $v(k)$ is the noise, $u(k)$ is the filter input, $y(k)$ is the filtered output, and $e(k)$ is the error. The presence of noise, $v(k)$, will cause the output $y(k)$ to be corrupted by transformed noise included in $v(k)$, namely random walk, white noise, etc. A typical goal of using an IIR filter is to minimize the MSE of the error $e(k)$ (e.g., the L-2 norm of the error). Detailed IIR filtering descriptions and definitions can be found in nearly all standard signal processing textbooks; for example, see Williams and Taylor (2006); Schlichthärle (2011).

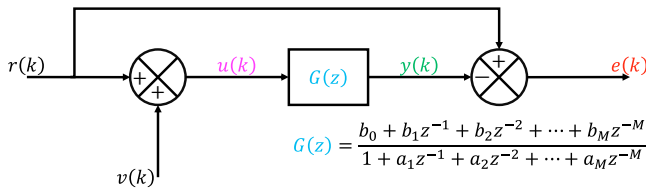


Fig. 1. System diagram of an IIR filter.

This work is organized as follows: Section 2 presents the AVAR extension applied to the output of IIR filters. Section 3 presents the use of the area of an AVAR curve to select an MSE-minimizing IIR filter. Section 4 presents the method to find an AVAR-equivalent FIR filter to approximate an IIR filter's response. Conclusions then summarize the main contributions.

2. INFINITE IMPULSE RESPONSE FILTER

Without loss of generality and because of their commonality, this work demonstrates AVAR analysis of IIR filters using low-pass Butterworth filters as an example. Two key assumptions are made: first, that the filter is stable, indicated by its eigenvalues being within the unit circle, and second, that the effect of initial conditions is negligible at samples higher than the order of the AVAR-equivalent FIR filter. Equation (1) presents the discrete implementation of an IIR filter with order M .

$$y_j^M = \sum_{i=0}^M b_i u_{j-i} - \sum_{i=1}^M a_i y_{j-i} \quad (1)$$

where y_j^M is the IIR filter output, u_{j-i} is the IIR filter input, and M is the IIR filter order.

The choice of initial conditions significantly influences the response of any IIR filter. The effect of the initial condition depends on both the initial condition and properties of the filter. In this work, the drift input starts from zero and all initial conditions are set to zero to minimize the effect of initial conditions on the response of the IIR filter. An illustration of the input (black dots) and output (magenta asterisks) of an IIR filter with order $M = 3$ and normalized cutoff frequency $\omega_n = 0.25$ is shown in Figure 2.

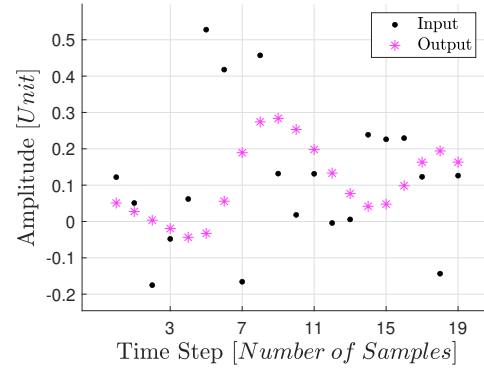


Fig. 2. I/O of an IIR filter with order $M = 3$ and normalized cutoff frequency $\omega_n = 0.25$.

2.1 Estimated AVAR of IIR filter output

In Maddipatla and Brennan (2024, 2025), the AVAR estimate of a signal presented in Allan (1987) is extended to the outputs of MA and FIR filters, respectively. In this section, equation (2) expands the AVAR estimation to the output of IIR filters.

$$\sigma_A^2[m] = \frac{1}{2(N-2m)} \sum_{l=2m+1}^N (\bar{y}_l^M - \bar{y}_{l-m}^M)^2 \quad (2)$$

where m is the correlation interval, and N is the data length. Furthermore, $\bar{y}_l^M$ is defined by equation (3).

$$\bar{y}_l^M = \frac{1}{m} \sum_{j=l-m}^{l-1} y_j^M \quad (3)$$

where y_j^M is the IIR filter output.

The reference input signal used in this study is drift (random walk coefficient of $0.02 \frac{\text{Unit}}{\sqrt{s}}$), and the noise is white noise (power spectral density of $0.0004 \frac{\text{Unit}^2}{s}$). Figure 3 illustrates the AVAR of the input (black) and the output (magenta) for a low-pass IIR filter with an order of $M = 3$ and a normalized cutoff frequency $\omega_n = 0.25$. Alongside the nominal AVAR curves (solid lines), the figure includes the 95th percentile lower bound (LB) and upper bound (UB) to indicate the statistical similarities/differences between the input and output AVAR curves, see Galleani (2011). It is observed that the low-pass IIR filter reduces the AVAR for correlation intervals of smaller than approximately 10 samples. However, the input and output curves appear statistically equivalent for higher correlation intervals, specifically those greater than 100 samples, an expected result because a low-pass IIR filter impacts only high-frequency content in the input signal.

Figure 4 demonstrates the effect of changing the normalized cutoff frequency on the AVAR of output using low-pass IIR filters with an order of $M = 3$. The results indicate that the AVAR decreases at shorter correlation intervals as the corner frequency decreases due to reduced high-frequency content in the output signal. In contrast, at larger correlation intervals, the AVAR remains unchanged because low-pass IIR filters do not affect the low-frequency content of the input signal.

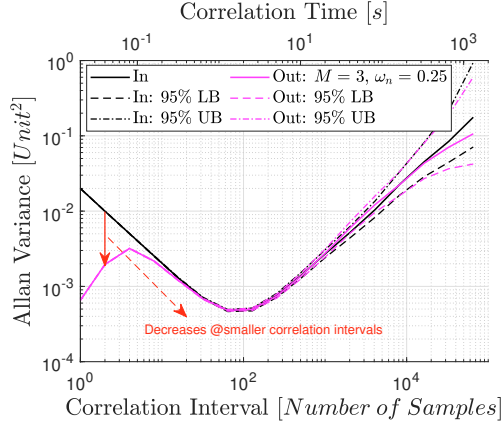


Fig. 3. AVAR of input (In) and output (Out) of an IIR filter with order $M = 3$ and normalized cutoff frequency $\omega_n = 0.25$. In addition to the nominal outputs (solid lines), the 95th percentile lower bound (LB) and upper bound (UB) are also shown.

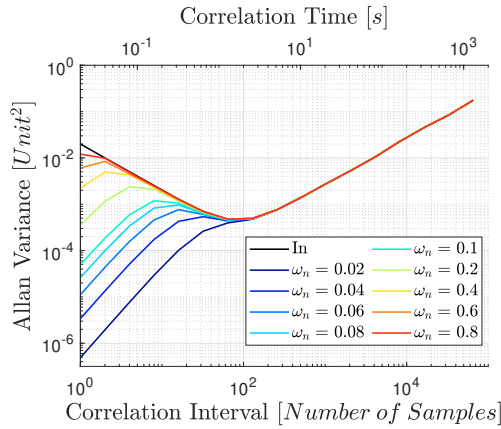


Fig. 4. Effect of normalized cutoff frequency ω_n on AVAR of the output of an IIR filter with order $M = 3$.

3. DRIFT INPUT CORRUPTED BY WHITE NOISE

In Maddipatla and Brennan (2024, 2025), the area under the AVAR curve of error is introduced as an efficient alternative measure to MSE for optimizing MA and FIR filters, respectively. Figure 5 compares the area under the AVAR curve of error against the MSE at instance $k = 524,287$ for IIR filters. MSE (dashed lines) resembles the area under the AVAR curve (solid lines) of error, suggesting that an MSE optimal IIR filter is similar to the filter minimizing area under the AVAR curve. The maximum order of the IIR filter examined in this context is capped at 10 since there is minimal variation in MSE with respect to the filter order. Like MA and FIR filters, the area under the AVAR curve offers a computational advantage over MSE in identifying an MSE optimal IIR filter. In this case, the area of AVAR requires only one iteration per filter. In contrast, MSE-based filter optimization requires approximately 500 iterations.

4. AVAR-EQUIVALENT FIR FILTER TO AN IIR

This section adapts AVAR to determine the order of the FIR filter that best approximates a given IIR filter. An

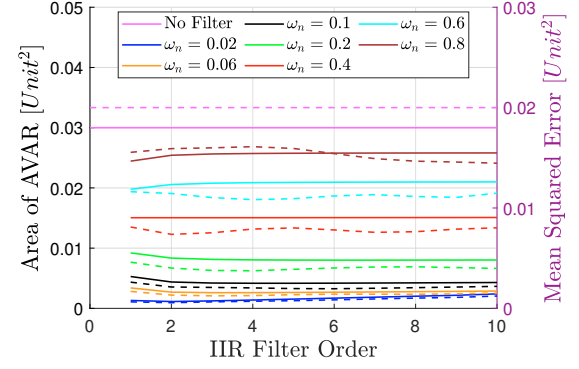


Fig. 5. The error analysis in an IIR filter using the MSE versus the area of AVAR. The MSE (dashed lines) calculation averaged over 500 iterations converges to the same optimal filter as the area of AVAR (solid lines) from 1 iteration.

FIR filter is considered to be approximating an IIR filter if both filters produce the same response for the same input, i.e., the AVAR of errors from a given IIR filter and its AVAR-equivalent FIR filter should be equal within a user-defined epsilon bound. IIR filters can be represented using two common methods: by transfer function and via state space. Specifically, IIR filters defined by equation (1) can also be represented in state-space form by equation (4).

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ y(k) &= Cx(k) + Du(k) \end{aligned} \quad (4)$$

Equation (5) provides the output for IIR filters represented in state-space form. The output has two major components: the response due to initial conditions and the response to the input. Regardless of the order of the IIR filter, the output at time index j depends on all the data up to that point.

$$y_j = \underbrace{CA^j x_0}_{\text{Initial Condition response, } y_j^{ic}} + \underbrace{Du_j + CBu_{j-1} + CABu_{j-2} + \dots + CA^{j-1}Bu_0}_{\text{Input response, } y_j^u} \quad (5)$$

In comparison, the output of an FIR filter with order p at time index j depends only on the most recent $p+1$ data points. One significant advantage of using FIR filters is that they require a fixed amount of data to estimate the output. In contrast, IIR filters necessitate an increasing data size. It is essential to note that FIR filters here are not defined by filter order and normalized cutoff frequency. Instead, an FIR filter of order p is defined by the first $p+1$ coefficients of the input response of the IIR filter, as indicated in equation (6).

$$\begin{aligned} y_j^u &= Du_j + CBu_{j-1} + CABu_{j-2} + \dots + \\ &\quad CA^{p-1}Bu_{j-p} + \dots + CA^{j-1}Bu_0 \\ &= Du_j + \underbrace{\sum_{i=1}^p CA^{i-1}Bu_{j-i}}_{\text{AVAR-Equivalent FIR response}} + \underbrace{\sum_{i=p+1}^j CA^{i-1}Bu_{j-i}}_{\text{IIR response}} \end{aligned} \quad (6)$$

The output of an AVAR-equivalent FIR filter of order p is defined by equation (7).

$$\begin{aligned}\tilde{y}_j^u &= Du_j + CBu_{j-1} + CABu_{j-2} + \dots + CA^{p-1}Bu_{j-p} \\ &= Du_j + \sum_{i=1}^p CA^{i-1}Bu_{j-i}\end{aligned}\quad (7)$$

The coefficients of an AVAR-equivalent FIR filter of order p are shown in equation (8).

$$c_i = \begin{cases} D & i = 0 \\ CA^{i-1}B & 1 \leq i \leq p \end{cases} \quad (8)$$

The objective of determining an AVAR-equivalent FIR filter is to estimate the order p of the FIR filter so that the difference between the AVAR responses of the FIR and IIR filters is minimized. A minimal difference in responses indicates a corresponding minimal difference in errors produced by both filters within an AVAR analysis. Thus, an FIR filter can be considered AVAR-equivalent to an IIR filter if the error from both filters have epsilon-bounded AVAR error curves. Mathematically, the order p of the equivalent FIR filter is estimated from equations (9) and (10). Specifically, for a p^{th} -order FIR filter to be AVAR-equivalent to an IIR filter, the nominal distance between the AVAR curves of errors from FIR filters of orders p and $p+1$ should be smaller than a user-defined value, as specified in equation (10).

$$\|e\|_2^p = \sqrt{\sum_m \left(\frac{\sigma_{\tilde{y}^{p+1}}^2[m]}{\sigma_{\tilde{y}^p}^2[m]} - 1 \right)^2} \quad (9)$$

$$\min_{\|e\|_2^p < \epsilon} p \quad (10)$$

The relative difference between two parameters is typically defined as the difference between the two parameters divided by one of them. Figure 6 illustrates the relative distance between two AVAR curves. The distance between two AVAR curves at a correlation interval is defined as the absolute difference in AVAR at that interval. The height of the AVAR curve at a given correlation interval corresponds to the AVAR at that specific interval. The relative distance between two AVAR curves at a particular correlation interval is calculated by dividing the distance between the curves with the height of the AVAR curve from the FIR filter with the lowest order at that interval. Furthermore, the overall relative distance between two AVAR curves is determined as the square root of the sum of the squared relative distance between AVAR at all correlations at which both AVAR curves are estimated.

Algorithm 1 is used to select the order of the AVAR-equivalent FIR filter to an IIR filter. Figure 7 demonstrates the algorithm using an IIR filter of order $M = 3$ and normalized cutoff frequency of $\omega_n = 0.25$. It is important to note that the relative distance between the AVAR curves of error defined by equation (9) decreases with increasing filter order, indicating that there is no optimal order for the AVAR-equivalent FIR filter. So, the order of the AVAR-equivalent FIR filter is selected based on equation (10), i.e., the minimum order of the filter, after which the relative distance never exceeds a user-specified tolerance, ϵ . In the example shown in Figure 7, the relative distance between AVAR curves is less than $\epsilon = 0.01$ and never exceeds that limit at an order higher than 23,

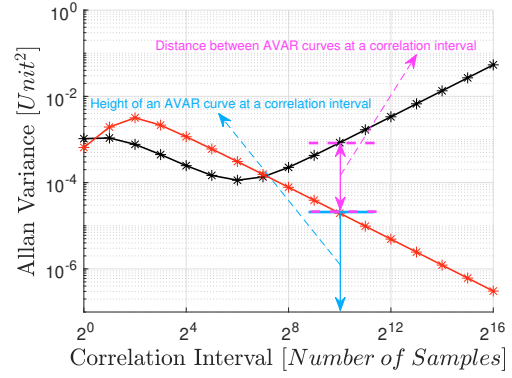


Fig. 6. The similarity between two AVAR curves is measured in this work as the distance between AVAR curves at a correlation interval divided by the height of one of the AVAR curves at the same correlation interval.

Algorithm 1 AVAR-Equivalent FIR order to IIR

- 1: Estimate a state-space form of the IIR filter.
 - 2: Choose the selection criteria, ϵ .
 - 3: Initialize the order of FIR filter, $p = 1$, the interim order of FIR, $inter_p = NaN$, and the AVAR-equivalent FIR selection status, $not_fir_selected = True$.
 - 4: **while** $not_fir_selected$ **do**
 - 5: Estimate coefficients of p^{th} -order FIR filter using equation (8).
 - 6: Estimate AVAR of error from an FIR filter derived in above step, see Maddipatla and Brennan (2025).
 - 7: **if** $p > 1$ **then**
 - 8: Estimate nominal distance $\|e\|_2^{p-1}$ between AVAR curves of errors using equation (9).
 - 9: **if** $\|e\|_2^{p-1} < \epsilon$ **then**
 - 10: **if** $inter_p$ is NaN **then**
 - 11: Set interim FIR order, $inter_p = p+50$.
 - 12: **else if** $p == inter_p$ **then**
 - 13: Set FIR order to $p-51$.
 - 14: Set $not_fir_selected$ to *False*.
 - 15: **end if**
 - 16: **else**
 - 17: Set interim FIR order, $inter_p = NaN$.
 - 18: **end if**
 - 19: **end if**
 - 20: $p \leftarrow p + 1$
 - 21: **end while**
-

implying that the order of the AVAR-equivalent FIR filter is $p = 23$.

The FIR filter selected based on the relative distance between the AVAR curves of error from FIR filters yields a response similar to that of the target IIR filter. Figures 8, 9, and 10 illustrate the process of selecting FIR filters that replicate the response of IIR filters using three different IIR filters. Specifically, Figure 8 showcases the selection of the FIR filter that corresponds to the IIR filter with an order of $M = 3$ and a normalized cutoff frequency of

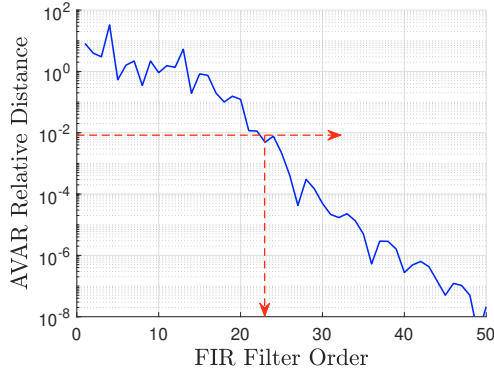


Fig. 7. FIR filter of order $p = 23$ is AVAR-equivalent to an IIR filter with order $M = 3$ and normalized cutoff frequency $\omega_n = 0.25$ for an $\epsilon = 0.01$

$\omega_n = 0.25$. Figure 8a shows the AVAR of the error from the IIR filter alongside the AVAR of the error from FIR filters of various orders. It can be observed that the AVAR of the FIR filter becomes statistically equivalent (equivalent within the uncertainty bounds of the AVAR curve) to that of the IIR filter with increasing filter order. Additionally, the relative distance between the AVAR curves of error from the FIR filters continues to decrease with increasing filter order, as shown in Figure 8b. At an FIR filter order of $p = 23$, the relative distance between the AVAR curves satisfies the selection criteria, meaning that the relative distance is smaller than $\epsilon = 0.01$. Figure 8c compares the response of the IIR filter (in red) against the response of the AVAR-equivalent FIR filter (in green). The responses of the AVAR-equivalent FIR filter and IIR filter are nearly indistinguishable visually.

Likewise, Figure 9 demonstrates the selection of the FIR filter AVAR-equivalent to the IIR filter, which has an order of $M = 3$ and a normalized cutoff frequency of $\omega_n = 0.02$. As seen in Figure 9a, the AVAR of the error from the FIR filter, with an order of $p = 226$, is statistically equivalent to that of the IIR filter. The findings in Figure 9b further corroborate this equivalence; the FIR filter with order $p = 226$ satisfies the selection criteria, i.e., the relative distance between the AVAR curves of error from FIR filters is less than $\epsilon = 0.01$. Also, Figure 9c reinforces that the response from the AVAR-equivalent FIR filter matches that of the IIR filter. Notably, as the normalized cutoff frequency decreased from 0.25 to 0.02, the order of the AVAR-equivalent FIR filters increased from 23 to 226.

Figure 10 depicts the selection of the order of the FIR filter that is AVAR-equivalent to the IIR filter, which has an order of $M = 5$ and a normalized cutoff frequency of $\omega_n = 0.02$. As demonstrated in the previous examples, the AVAR curve of error from the FIR filter satisfying the selection criteria defined by equation (10) with $\epsilon = 0.01$ is statistically equivalent to the AVAR of the error from the IIR filter, as seen in Figure 10a. Moreover, the response of the AVAR-equivalent FIR filter with an order of $p = 352$ is visually indistinguishable from that of the IIR response, as shown in Figure 10c. Furthermore, the order of the AVAR-equivalent FIR filters rose from 226 to 352 when the order of the IIR filter was raised from 3 to 5. It is important to note that the relationship between the order, normalized cutoff frequency of the IIR filter

and the order of the AVAR-equivalent FIR filter is not straightforward, as the order of FIR filters is primarily influenced by the eigenvalues, which are affected by the filter order, normalized cutoff frequency, and the specific type of IIR filters.

5. CONCLUSIONS

This work extends AVAR methods to IIR filters, leading to the following conclusions: 1) Similar to MA and FIR filters, the area under the AVAR curve of error can be leveraged to find an optimal IIR filter that responds to drift input corrupted by white noise. 2) The response of a stable IIR filter at any given instant does not necessarily depend on all the prior inputs. Instead, the relative distance between AVAR curves of error from FIR filters can serve as a metric for determining the minimal number of prior inputs needed—referred to as the order of the AVAR-equivalent FIR filter in this work—to estimate the IIR response accurately. The choice of selection criteria (ϵ) is based on data, and this work can be extended to choose it such that it relates to the confidence on equivalence between the IIR and FIR's output.

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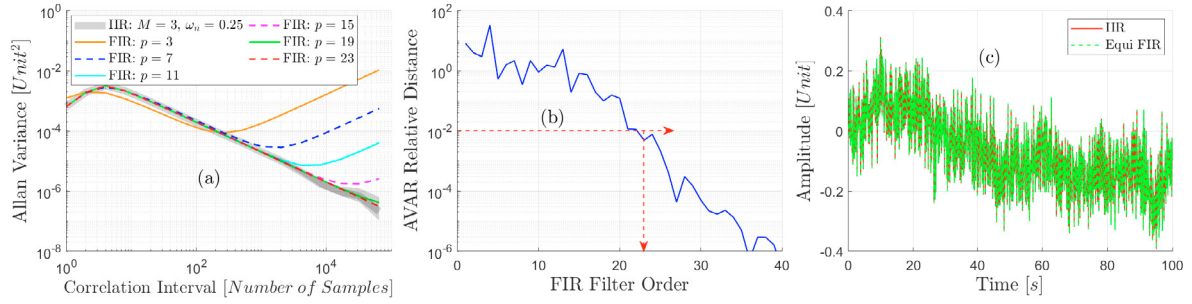


Fig. 8. FIR filter of order $p = 23$ is AVAR-equivalent to an IIR filter with order $M = 3$ and $\omega_n = 0.25$ for an $\epsilon = 0.01$.

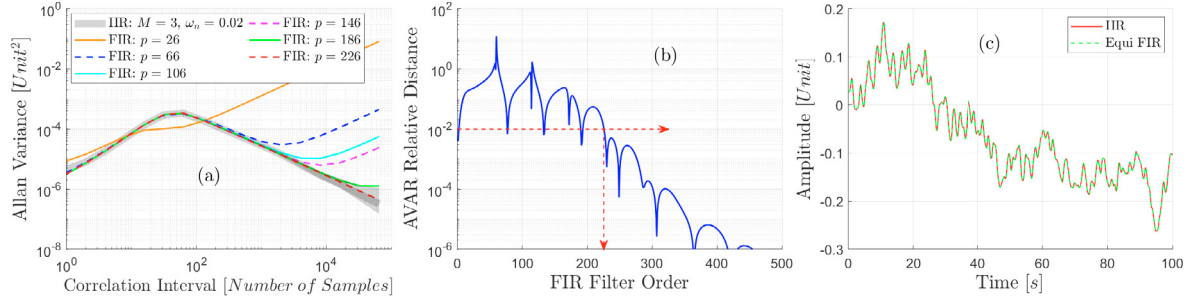


Fig. 9. FIR filter of order $p = 226$ is AVAR-equivalent to an IIR filter with order $M = 3$ and $\omega_n = 0.02$ for an $\epsilon = 0.01$.

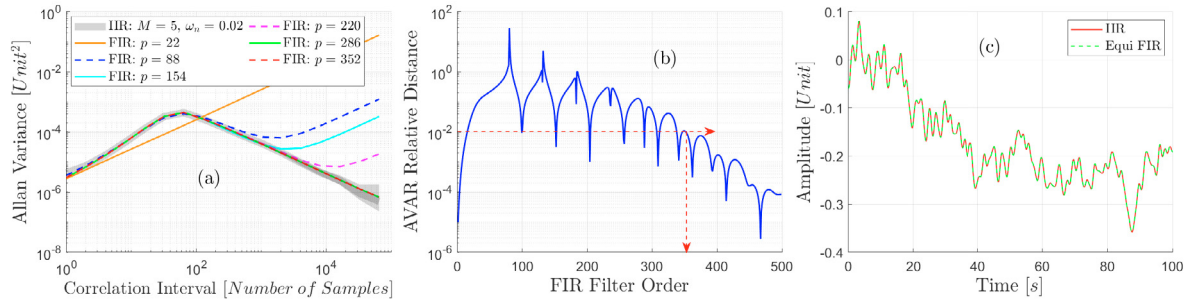


Fig. 10. FIR filter of order $p = 352$ is AVAR-equivalent to an IIR filter with order $M = 5$ and $\omega_n = 0.02$ for an $\epsilon = 0.01$.

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